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MOF Instability and Polymer Infiltration in Amine-Functionalized UiO MOFs/PIM-COOH-Based MMMs for Solvent-Based Separations

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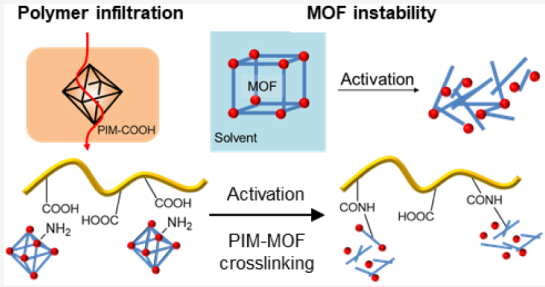
ABSTRACT: Organic solvent-based separations with membranes hold promise for reducing the energy demands of traditional thermally driven processes. However, their widespread adoption is hindered by the limited availability of commercial membranes and challenges related to their long-term stability and molecular weight selectivity. Mixed-matrix membranes (MMMs), which incorporate fillers such as metal−organic frameworks (MOFs) with well-defined pore structures, offer a pathway to improve selectivity while maintaining high throughput. Despite this promising opportunity, challenges with MOF stability and nonideal polymer−MOF interfaces persist and remain understudied for organic solvent nanofiltration (OSN) MMMs. In this study, MMMs using carboxylic acid-functionalized PIM-1 (PIM-COOH) and amine-functionalized UiO MOFs with varying pore sizes (UiO-6*x*-NH_2, *x* = 6, 7, 8) were developed. The functional groups allowed for facile polymer−MOF cross-linking to ensure membrane structural integrity. However, larger-pore MOFs faced stability issues, with the UiO-68-NH_2 framework collapsing when activated from solvents that had a high affinity to the MOF or solvents that had a high surface tension. Additionally, the UiO-67-NH_2 framework lost some of its crystalline character when cross-linked to PIM-COOH. The OSN performance of the MMMs was evaluated through solvent permeation and dye rejection. While the incorporation of UiO-66-NH_2 into PIM-COOH increased solvent permeability (a 36−143% increase compared to PIM-COOH at 21 wt % loading), the performance of MMMs incorporating larger-pore MOFs was hindered by MOF collapse and nonideal polymer−MOF interfaces that resulted in polymer infiltration, leading to up to a 78% decrease in permeability. Polymer infiltration effects were further investigated through CO_2 sorption and solid-state NMR experiments. This work underscores the critical challenges in the design of OSN MMMs, providing insights to guide the design of robust, high-performing materials for energy-efficient separations.



1. INTRODUCTION

Membrane-based separations present a significant opportunity to reduce the high energy consumption associated with traditional thermal-based separation methods, such as distillation, which currently account for more than 50% of the energy consumed in the industrial thermal separations sector.[1] While membrane technology has been widely adopted for water treatment, its application in organic solvent-based separation systems remains limited due to a lack of materials that can maintain stability and performance in harsh solvent environments. Organic solvent separations span various industries, from petrochemical to pharmaceutical, presenting substantial potential for energy savings if membrane technology can be effectively deployed.[2,3]

Solvent-based separations can be broadly categorized based on the size difference between the species being separated. Organic solvent nanofiltration (OSN) refers to the separation of large molecules, typically with molecular weights ranging from 200 to 1000 g mol^{-1}, from a solvent.[2] Applications of OSN include the recovery of catalysts, concentration and purification of active pharmaceutical ingredients, and processing of food oils.[4] On the other hand, organic solvent reverse osmosis (OSRO) involves solvent/solvent separations. One of the most important applications of OSRO is in oil refining.[2] Due to the smaller size difference between the species being separated, OSRO requires tighter control of pore size distribution, making it a more challenging separation compared to OSN.

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Most commercially available membranes are made from polymers due to their scalability and processability into various membrane form factors. However, polymers often experience swelling or complete dissolution in solvent environments, thus altering their size selectivity[5] and structural integrity. Consequently, designing membranes with precise size-sieving capabilities remains a primary challenge for solvent-based separation systems. To enhance the size selectivity in liquid-based separation membranes, researchers have incorporated additives with precise and tunable pore sizes into the membranes. For example, Tu et al. integrated membrane proteins with diameters ranging from 0.8 to 1.5 nm into poly(butadiene)-*b*-poly(ethylene oxide), a block copolymer.[6] The membrane proteins serve as water channels, yielding 1.5−45 times higher water permeability than the state-of-the-art membranes reported in the literature while maintaining similar rejection profiles. Similarly, Corcos et al. demonstrated tunable salt rejection by adjusting the pore size of covalent organic frameworks (COFs) from 3.2 to 3.4 nm in their composite membranes.[7] Jiang et al. incorporated aligned macrocycle pores with various cavity sizes to achieve tunable cutoff rejections (≥90%) for dye molecules from ∼0.45 to 0.8 nm while maintaining high solvent throughput.[8]

Metal−organic frameworks (MOFs) are a promising class of materials for separation applications. MOFs are composed of metal clusters bonded to organic linkers, offering a wide range of chemistries and structures.[9] This variety allows for the enhancement of species-specific affinity through functionalization and provides precise control over a wide range of pore sizes. However, their crystalline nature makes it difficult for MOFs to form defect-free membranes. Mixed-matrix membranes (MMMs) have emerged as a compelling strategy to combine the processability of polymers with the advantageous properties of MOFs, making them an area of significant research interest.[9] Studies in the gas separation field have shown that the incorporation of MOFs into membranes introduces additional pores and subsequently boosts gas permeability.[10] The well-defined porous nature of MOFs enhances diffusion selectivity, while their chemical versatility allows for improved sorption selectivity through chemical functionalization.[11−13] Moreover, the rigid crystalline structure of MOFs often limits polymer chain mobility, thus increasing resistance to phenomena that often preclude the use of polymer membranes in industrial gas separations such as plasticization and physical aging.[14,15]

The benefits observed in gas separations have led to interest in using MMMs for solvent-based separations. Among the MOFs used for these applications, UiO (University of Oslo) MOFs are particularly prominent. UiO-66, composed of $Zr_6O_4(OH)_4$ nodes bridged by 1,4-benzenedicarboxylate (bdc) linkers, is one of the most extensively studied MOFs due to its well-established synthesis procedure and stability in solvents.[16] While all UiO-based MOFs share the same metal node structure, they can incorporate a variety of functional linkers, making them a versatile platform with diverse functional groups and pore structures. UiO-67 and UiO-68 are isoreticular variants of UiO-66, formed by replacing the bdc linker with the longer 4,4′-biphenyldicarboxylate (bpdc) or terphenyl-4,4′-dicarboxylate (tpdc), respectively, resulting in larger pore apertures.[17] These UiO MOFs are particularly relevant for OSN due to their pore sizes aligning with the OSN molecular separation range.[2] Notably, Cseri et al. demonstrated a tunable molecular weight cutoff (MWCO) for dyes at 90% rejection, ranging from ∼150 to 350 g mol^{-1}, by incorporating amine-functionalized UiO MOFs with different pore apertures (UiO-6*x*-NH_2, where *x* = 6, 7, or 8) into a polybenzimidazole matrix. This work highlights the potential of UiO MOF-based MMMs in solvent-based separations.[18]

Despite the advantages of MOF-based MMMs, major challenges persist in their preparation. Issues such as polymer−MOF incompatibility and MOF particle agglomeration can result in defects in the membranes, leading to nonselective transport and low mechanical integrity.[19] In liquid-based separations, poor polymer−MOF interaction is especially detrimental because it can cause MOF particles to leach out of the MMM and further contaminate the product stream. Common strategies to prepare MMMs with good polymer−MOF interactions for solvent-based separations include in situ growth of MOFs in a polymer,[20] tethering polymer chains onto MOFs,[18] embedding MOFs into a polymer cross-linked network via interfacial polymerization,[21,22] and directly cross-linking MOFs to the polymer.[23] In addition to interfacial incompatibility, other issues contribute to hindered MMM performance such as polymer infiltration into MOF pores, which lowers membrane permeability, and MOF structural instability especially during activation. While challenges with MMM design and MOF activation have been studied extensively for gas separations[24,25] and pure MOFs,[26,27] respectively, they remain relatively unexplored for liquid-based applications.

In this study, MMMs were prepared from UiO-6*x*-NH_2 (*x* = 6, 7, or 8) and a carboxylic acid-functionalized PIM-1 (PIM-COOH).[28] Our previous study on gas separations showed that the good interfacial compatibility between UiO-66-NH_2 and PIM-COOH improved plasticization resistance, as evidenced by the increase in CO_2 plasticization pressure from 10 atm to beyond 40 atm for a 35 vol % UiO-66-NH_2/PIM-COOH MMM compared to pure PIM-COOH films,[29] due to the hydrogen bonding between the $-NH_2$ in the MOF and the −COOH in PIM-COOH. Extending this platform to solvent-based applications is advantageous, as it allows for facile cross-linking between the amine groups on the MOFs and the carboxylic acid groups on the polymer via a carbodiimide cross-linker. The UiO-6*x*-NH_2 MOFs were successfully prepared, and challenges with handling larger-pore MOFs were identified. UiO-6*x*-NH_2/PIM-COOH thin-film MMMs were fabricated via spin-casting, followed by a cross-linking reaction to ensure robust structural integrity. The effects of MOF incorporation on the membrane's performance in OSN were evaluated through pure solvent permeation and dye rejection tests in methanol solutions. Polymer infiltration leading to MOF pore blockage was observed in all MMMs, as indicated by CO_2 sorption measurements. This phenomenon was further investigated using solid-state cross-polarization magic angle spinning (SS CP-MAS) nuclear magnetic resonance (NMR). This systematic and detailed study reveals challenges and opportunities to form polymer−MOF hybrid materials for applications in the OSN.

2. RESULTS AND DISCUSSION

2.1. Material Synthesis and Characterization. The synthesis of the PIM-COOH polymer was carried out by hydrolyzing the nitrile groups in PIM-1 into carboxylic acid groups using sulfuric acid, following previously reported procedures.[29] The thermogravimetric analysis (TGA) profile of PIM-COOH in air shows a distinctive weight loss due to

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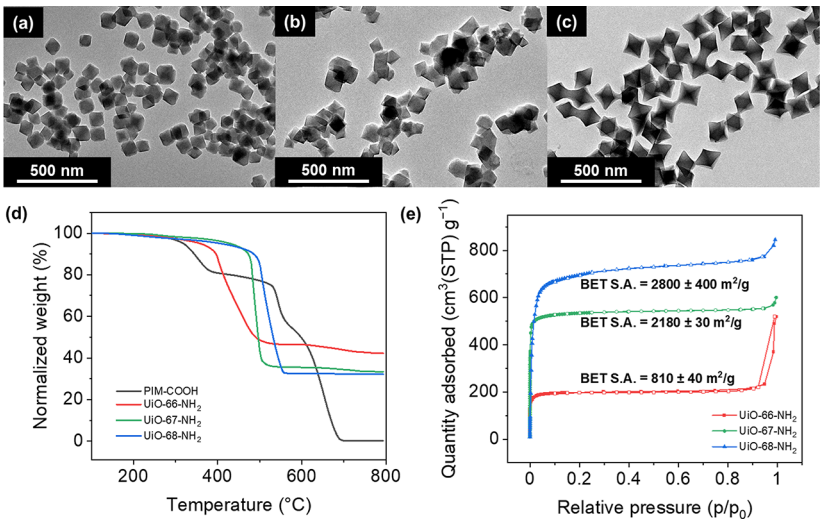

Figure 1. TEM images of (a) UiO-66-NH_2, (b) UiO-67-NH_2, and (c) UiO-68-NH_2 particles used for thin film preparation in this study. The particle sizes of the samples in (a−c) are 97 ± 14 nm, 91 ± 22, and 126 ± 13 nm, respectively. (d) TGA profiles in air for PIM-COOH, UiO-66-NH_2, UiO-67-NH_2, and UiO-68-NH_2 (activated from hexane). All TGA samples were held at 120 °C for 30 min to remove any adsorbed water and then ramped from 100 to 800 °C at 10 °C min^{-1}. The profile was normalized to the weight at 100 °C at the beginning of the ramping step. (e) N_2 adsorption/desorption isotherms at 77 K for UiO-66-NH_2, UiO-67-NH_2, and UiO-68-NH_2. Filled and open symbols represent adsorption and desorption, respectively. For BET measurements, UiO-66-NH_2 and UiO-67-NH_2 were activated from methanol, whereas UiO-68-NH_2 was activated from hexane.

decarboxylation occurring between 300 and 400 °C (Figure 1d). The 14.6% mass loss in this temperature range is due to the loss of the −COOH functionality.[30] This mass loss is compared to the theoretical loss of 17.7% at full conversion. The linkers for UiO-67-NH_2 and UiO-68-NH_2 were synthesized via a Suzuki−Miyaura reaction based on a reported method,[18] while the bdc-NH_2 linker for UiO-66-NH_2 was obtained commercially. Detailed synthesis procedures for 2-amino-[1,1'-biphenyl]-4,4'-dicarboxylic acid (bpdc-NH_2) and 2'-amino-[1,1':4',1''-terphenyl]-4,4''-dicarboxylic acid (tpdc-NH_2) are provided in Sections S1.2 and S1.3. The chemical structures of the synthesized linkers were characterized by Fourier-transform infrared spectroscopy (FT-IR), ^{1}H NMR, and ^{13}C NMR, as shown in Figures S4−S9. All additional details on materials, methods, and characterization can be found in the Supporting Information.

UiO-6*x*-NH_2 MOF particles were synthesized via an acetic acid-modulated solvothermal synthesis. Details on the synthesis procedure for UiO-66-NH_2 have been reported previously,[29] and details on the synthesis procedures for UiO-67-NH_2 and UiO-68-NH_2 are provided in Sections S1.4 and S1.6, respectively. To ensure good dispersion and optimal contact with the polymer, nanosized MOF particles were prepared.[24] For a consistent comparison of MMMs made from different MOFs, the particle sizes needed to be uniform across all MOF types considered. To best match the particle size of UiO-66-NH_2 particles synthesized (97 ± 14 nm), various synthetic parameters for UiO-67-NH_2 and UiO-68-NH_2 were adjusted. Specifically, a systematic study was conducted to investigate the effects of linker equivalence, acetic acid equivalence, water equivalence, and reaction concentration on the synthesized UiO-67-NH_2 particle size. The study revealed that smaller particle sizes could be achieved with higher linker equivalence, lower acetic acid equivalence, and higher water equivalence, aligning with the expected MOF nucleation-crystal growth mechanisms.[31] Higher linker equivalence promotes linker-metal bond formation, depleting the local metal concentration in the reaction and restricting further growth. Increased modulator equivalence competes with linkers for coordination sites, slowing the crystal growth. Water facilitates the formation of the $Zr_6O_4(OH)_4^{12+}$ cluster and reduces particle size at high concentrations, as reported in the literature.[32,33] Reaction concentration did not significantly affect particle size, although it was found that concentrations below 3 mg of $ZrCl_4$/mL DMF did not produce any crystalline product. More details on the range of parameters explored and the resulting UiO-67-NH_2 particle sizes can be found in Section S1.5 and Figure S1. Among the parameters studied, water equivalence had the most significant impact on particle size without sacrificing the yield. The guidelines developed for UiO-67-NH_2 were successfully applied to control the size of UiO-68-NH_2. Using this strategy, UiO-6*x*-NH_2 MOFs with particle sizes around 100 nm were prepared, as shown in the transmission electron microscopy (TEM) images in Figure 1a−c.

The thermal stability, crystallinity, and Brunauer−Emmett−Teller (BET) surface area of the MOF particles were characterized by TGA, powder X-ray diffraction (PXRD), and N_2 adsorption at 77 K, respectively. The TGA profiles in air (Figure 1d) show that all MOFs exhibit a gradual mass loss between 100 and 350 °C, corresponding to the removal of residual solvents and monocarboxylate linkers, as well as the dehydroxylation of Zr_6. This first stage of mass loss is followed by a drastic mass loss starting at ∼400−500 °C due to the framework decomposition into ZrO_2.[34,35] The residual mass at 800 °C for each MOF and polymer was subsequently used to quantify the MOF loading in the MMMs. Additional details on the TGA data collection are provided in Section S2.5.

For PXRD and N_2 sorption experiments, MOF particles were initially activated by centrifugation from methanol, followed by drying under vacuum at 100 °C overnight. UiO-66-NH_2 and UiO-67-NH_2 were successfully activated, as indicated by the PXRD patterns in Figure S10a and the BET surface area measurements in Figure 1e. The BET surface areas

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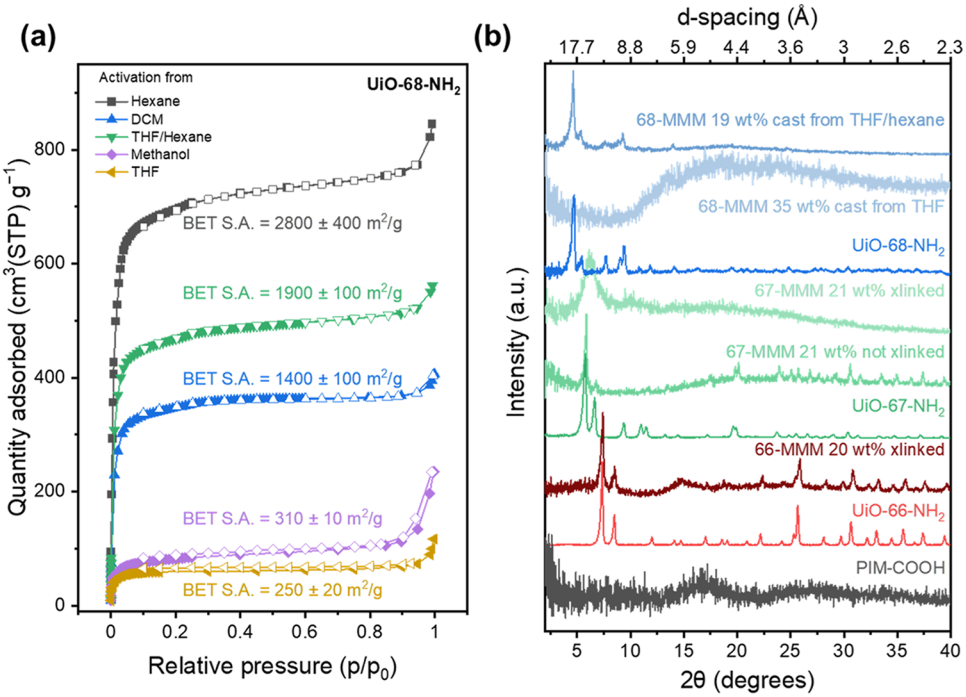

Figure 2. (a) N_2 adsorption/desorption curves of the UiO-68-NH_2 powders activated from various solvents. Filled and open symbols represent adsorption and desorption, respectively. The fitted BET surface area of each sample is indicated in the plot. (b) PXRD patterns of PIM-COOH, UiO-6*x*-NH_2 MOFs, and PIM-COOH/UiO-6*x*-NH_2 MMMs.

obtained for UiO-66-NH_2 and UiO-67-NH_2 are comparable to those reported in the literature. Examples of reported values are 876 m^2 g^{-1} for UiO-66-NH_2[36] and 2180 m^2 g^{-1} for UiO-67-NH_2.[37] It is worth noting that BET surface areas for the same MOF can vary significantly due to the differences in defect ratios arising from different synthesis procedures,[33] but these effects are not considered in this study.

In contrast, the removal of methanol from the UiO-68-NH_2 lattice led to a structural collapse. The pore diameters, measured by the largest hypothetical sphere that could fit in each framework, were found to be 6.20 Å for UiO-66-NH_2, 11.67 Å for UiO-67-NH_2, and 14.55 Å for UiO-68-NH_2 (cf. Table S4). MOFs with larger pore sizes are more prone to structural collapse during solvent removal because larger pores allow solvent molecules to cluster and exert stronger capillary stresses.[26] To mitigate this unwanted effect, solvents with lower boiling points and surface tensions are preferred.[26,27,38] Figures S10b and 2a compare the PXRD patterns and BET surface areas of the UiO-68-NH_2 dried from various solvents. The surface tensions of these solvents are listed in Table S1. Hexane, with the lowest surface tension, preserved the highest BET surface area, while solvents like methanol and tetrahydrofuran (THF), which can form hydrogen bonds with the MOF, induced structural collapse. These findings underscore the importance of selecting appropriate solvents to maintain the structural integrity of MOFs during activation.

2.2. MMM Preparation and Characterizations. MMMs consisting of PIM-COOH and UiO-6*x*-NH_2 (denoted as 6*x*-MMMs) were fabricated into freestanding bulk films and thin films. The bulk films were used for PXRD and CO_2 sorption measurements, while the thin films were used for solvent permeation testing. The MOF loading in the MMMs was determined by TGA and is reported as wt % throughout the study. The 66- and 67-MMMs were prepared by forming a well-dispersed MOF suspension in a viscous PIM-COOH solution in THF. The mixture was then either cast in a Petri dish to form bulk films or spin-cast onto a commercial polyacrylonitrile (PAN) substrate to form thin films (Figure S11). Given that UiO-68-NH_2 loses its crystallinity during THF removal and that solvent removal is inevitable during both spin-casting and evaporative casting, a cosolvent of 26.8 wt % hexane in THF was used to prepare 68-MMMs. This cosolvent concentration was chosen because it is the highest concentration of hexane in THF that can dissolve approximately 6 wt % PIM-COOH.

To ensure the structural integrity of the MMMs in solvent environments, PIM-COOH was cross-linked to the MOFs using EDC/NHS (1-ethyl-3-(3-(dimethylamino)propyl)-carbodiimide/*N*-hydroxysuccinimide) chemistry, which forms amide bonds by reacting the carboxylic acid groups with the amine groups. The cross-linking reaction was performed on the as-cast films. Detailed procedures for MMM preparation and cross-linking are given in Sections S1.7−S1.9. Polymer−MOF cross-linking was confirmed through NMR, film dissolution tests, and TEM images of the reacted MOF particles, as detailed in Section S3.4.

The crystallinity of the incorporated MOFs was assessed through PXRD measurements of the MMM bulk films. Figure 2b shows that the pristine PIM-COOH polymer is amorphous, while all the pristine MOFs show strong crystalline peaks. MOF crystallinity was retained in the 66-MMM after the cross-linking reaction, with a broad PIM-COOH peak also observed at around a 2θ value of 15 degrees. The 67-MMM preserved MOF crystallinity before the cross-linking reaction, but UiO-67-NH_2 partially lost its crystallinity during cross-linking and the subsequent drying step, leaving only a broad MOF peak at a 2θ value of 6 degrees. To further investigate this loss of crystallinity, the MOF stability in EDC/NHS solution was tested by stirring the MOFs in an EDC/NHS methanol solution overnight. TEM images and PXRD patterns of the

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treated MOF showed that MOF particles retained their structure and crystallinity (Figure S14a−c,e), suggesting that the observed crystallinity loss in cross-linked MMMs is due to the strong interaction between PIM-COOH and UiO-67-NH_2.

MOF crystallinity was preserved in the 68-MMM cast from the THF/hexane cosolvent, as shown by the PXRD profile. The profile of a 68-MMM cast from THF is also shown for comparison. The complete loss of MOF signals in the 68-MMM sample cast from THF underscores the importance of matching suitable MOF activation solvents with the solvents used for the MMM polymer matrix. It is worth noting that the 68-MMMs tested for PXRD were not cross-linked because cross-linking was performed in a methanol solution and drying the 68-MMM from methanol would result in the loss of MOF crystallinity. Furthermore, any solvent exchange steps on the cross-linked 68-MMMs could cause intense polymer swelling or contraction, potentially damaging the MOF structure. Clearly, these findings demonstrate the care that must be taken when forming MMMs with extended-structure MOFs.

Given the challenges of activating larger-pore MOFs, alternative methods for thin-film preparation may hold more promise. A notable study conducted by Cseri et al. fabricated MMMs using a phase inversion method, which does not involve solvent evaporation from the MOF during fabrication, thereby minimizing capillary stress and preserving structural integrity.[18] However, given the manufacturing and scale-up challenges in deriving clear structure−property relationships for phase-inverted suspension and due to the limited scale of PIM-COOH synthesis and the large amount of MOF required to develop a new MMM phase inversion approach, we instead focused on a materials-lean approach for casting and adopted a spin-casting technique.

2.3. Organic Solvent Nanofiltration (OSN) Performance. The organic solvent nanofiltration (OSN) performance of the prepared thin films was evaluated through pure-solvent permeation and dye rejection tests by using a dead-end filtration cell. The solvents tested included methanol (MeOH), isopropyl alcohol (IPA), acetonitrile (ACN), acetone, toluene, and heptane, covering a range of solubility parameters common for solvent testing. Solvent permeabilities were calculated by normalizing the flux with the transmembrane pressure times the thickness of the active layer (eq S3). Dye solutions were prepared in methanol at 32 ppm, with dyes having molecular weights ranging from 249 to 1017 g mol^{-1}, allowing for the determination of the MWCO at 90% rejection for all of the thin films tested. Detailed methods of the permeation tests are provided in Section S2.7.

Figures 3a and S15a plot the permeability and permeance, respectively, against the Hildebrand solubility parameters of solvents for PIM-COOH and 66-MMMs. The Hildebrand solubility parameter, defined as the square root of the cohesive energy density, is commonly used to determine the miscibility among substances, including liquids and polymers.[39,40] Among the solvents, acetonitrile and acetone exhibited the highest permeabilities, likely due to the higher swelling degree of PIM-COOH. PIM-COOH has a Hildebrand solubility parameter of 21.5 $MPa^{0.5}$, calculated from the group contribution method.[40] Therefore, PIM-COOH likely swells more in solvents with similar solubility parameters, leading to a higher solvent permeability. Although IPA also has a solubility parameter similar to that of PIM-COOH, its significantly higher viscosity (2.04 mPa·s, compared to less than 0.6 mPa·s for the other solvents tested)[41] leads to potentially reduced partitioning and

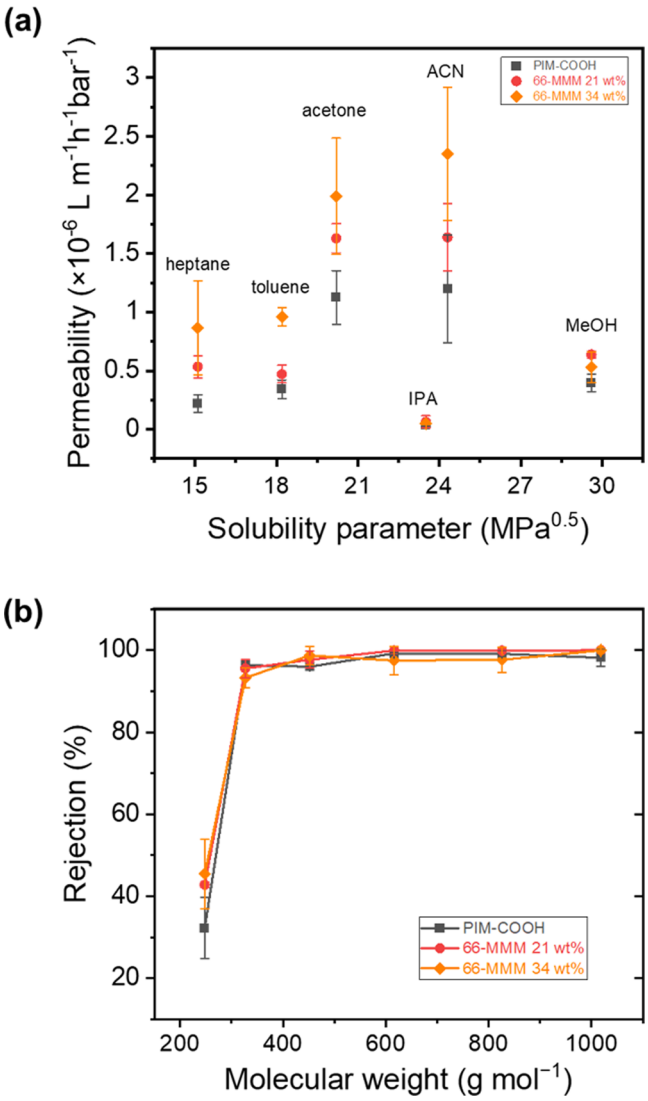

Figure 3. (a) Permeability of solvents and (b) rejection of dyes in methanol for PIM-COOH, a 21 wt % 66-MMM, and a 34 wt % 66-MMM. Permeance values of the same samples are plotted in Figure S15a.

a lower permeability. While the viscosity of bulk liquids does not directly describe the mobility of liquid in a dense membrane with a MWCO of less than ∼200 g mol^{-1}, Darvishmanesh et al. identified viscosity as an important parameter in considering the transport of liquids in microporous nanofiltration membranes, such as those used in this study.[42]

The effects of MOF loading on the performance of the OSN were studied using 66-MMMs. Figure 3a shows that increasing UiO-66-NH_2 loadings in MMMs enhanced the solvent permeability for all tested solvents. The increase in permeability is expected as MOF incorporation increases the free volume of the MMM. On the other hand, the rejection profiles of PIM-COOH and the 66-MMMs were found to be similar, as shown in Figure 3b. This result suggests that the pore size distribution of 66-MMM closely matches that of the pure PIM-COOH polymer in the swollen state. To underscore the importance of strong polymer−MOF interactions in MMMs for OSN, a non-cross-linked 66-MMM was tested as a control. Figure S13 shows that the rejection is generally lower

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and more variable when the 66-MMM is not cross-linked, highlighting the necessity of cross-linking to tighten and stabilize the MMM for OSN applications.

The effects of MOF pore sizes on permeation were investigated by comparing 66-MMM, 67-MMM, and 68-MMM. UiO-67-NH_2 and UiO-68-NH_2 have higher porosity and larger pore sizes than UiO-66-NH_2 (Figure 1e and Table S4), and thus, it would be expected that 67-MMM and 68-MMM would exhibit higher permeabilities than 66-MMM. However, Figure 4a shows that 67-MMM had lower solvent permeability in general compared to 66-MMM, and 68-MMM cast from THF and the THF/hexane cosolvent both showed lower permeabilities than the pristine PIM-COOH polymer. The lower permeability observed for MMMs incorporating larger-pore MOFs could be attributed to several factors: (1) the MOF structure may collapse during membrane fabrication; (2) polymer chains may rigidify at the polymer−MOF interface; and/or (3) polymer chains may infiltrate the MOF pores and limit pore accessibility. Each of these points will be described in detail later.

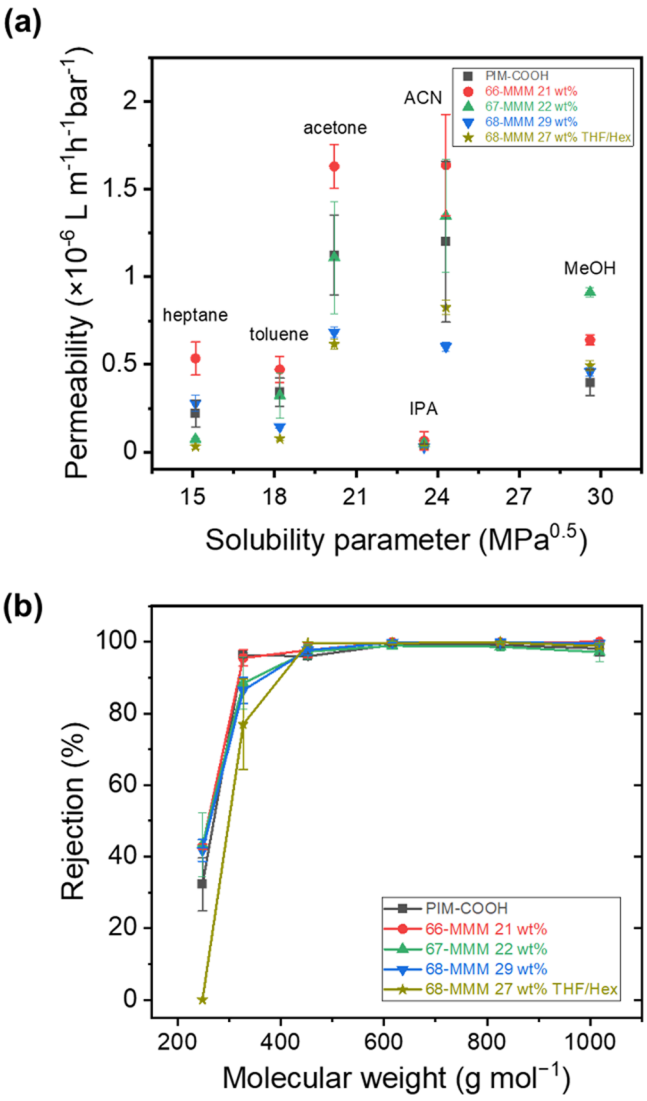

Figure 4. (a) Permeability of solvents and (b) rejection of dyes in methanol for PIM-COOH, a 21 wt % 66-MMM, a 22 wt % 67-MMM, a 29 wt % 68-MMM spin-cast from THF, and a 27 wt % 68-MMM spin-cast from the THF/hexane cosolvent. Permeance values of the same samples are plotted in Figure S15b.

In terms of the 67-MMM, the lower permeability in the MMM in relation to the polymer can be attributed to partial structural collapse during the cross-linking step, as supported by the PXRD results in Figure 2b. While 68-MMM preserved crystallinity when cast from THF/hexane, structural collapse during polymer−MOF cross-linking likely occurred. However, the crystallinity of the cross-linked 68-MMM was not checked because it would be difficult to decouple the effects, using PXRD measurements alone, of partial structural collapse due to either cross-linking or activation of the MOF from methanol. Nevertheless, we speculate that the same instabilities to cross-linking in UiO-67-NH_2 would be present in the less structurally stable UiO-68-NH_2 analog.

Polymer rigidification at the polymer−MOF interface is unlikely to be the cause of lower permeability, because MMM cross-linking was performed in methanol, which increases the polymer segmental motion, and the films were never completely dried after cross-linking. Thus, MMMs did not experience the vitrification process required for stress-induced rigidification.[43] Moreover, if significant rigidification were occurring, we anticipate that 66-MMM would also exhibit lower permeability than PIM-COOH. Given that all particles were synthesized to be approximately the same size, the polymer−MOF contact areas are comparable across all MMMs. Pore infiltration is a common reason for low permeability in MMMs, and larger-pore MOFs are more susceptible to this issue.[44] Pore blockage due to polymer infiltration was investigated through CO_2 sorption and solid-state NMR, as will be discussed later.

With the incorporation of larger-pore MOFs, an increase in the membrane MWCO is expected.[18] However, Figure 4b shows that there is no significant difference in the rejection profiles of 66-MMM, 67-MMM, and 68-MMM. MOF collapse and polymer infiltration prevent MMMs from accessing the full potential of well-defined MOF pore structures on separations. Consequently, the MWCO values obtained here do not accurately reflect the intrinsic OSN properties of UiO-6*x*-NH_2 MOFs with preserved structural features. The OSN performance of the MMMs in methanol is compared to literature values in Table S5.

To better understand MOF stability as a function of linker length and processing conditions, an in-depth study was performed to evaluate accessible MOF porosity and polymer−MOF interactions in the pure MOFs and MMMs when treated under various conditions. To analyze accessible MOF porosity, CO_2 sorption isotherms at 35 °C were collected for PIM-COOH, UiO-6*x*-NH_2 powders, and the MMMs, as shown in Figure 5a. The UiO-6*x*-NH_2 MOFs showed higher CO_2 uptake than PIM-COOH, which is expected due to their higher porosities. Notably, UiO-67-NH_2 and UiO-68-NH_2 had lower CO_2 sorption compared to UiO-66-NH_2 despite their higher BET surface areas (Figure 1e). This discrepancy is partially attributed to the higher amine group density present in UiO-66-NH_2. Similar findings have been reported elsewhere, where UiO-66-NH_2 showed higher CO_2 uptake than UiO-66 despite having a lower BET surface area.[45] All of the MMMs showed lower CO_2 uptake compared to their parent materials, suggesting that there was also pore blockage, even for the MMM incorporating the MOF with the smallest pore size, UiO-66-NH_2. Note that 66-MMM and 67-MMM were cross-linked, while 68-MMM was not. Therefore, the lower CO_2

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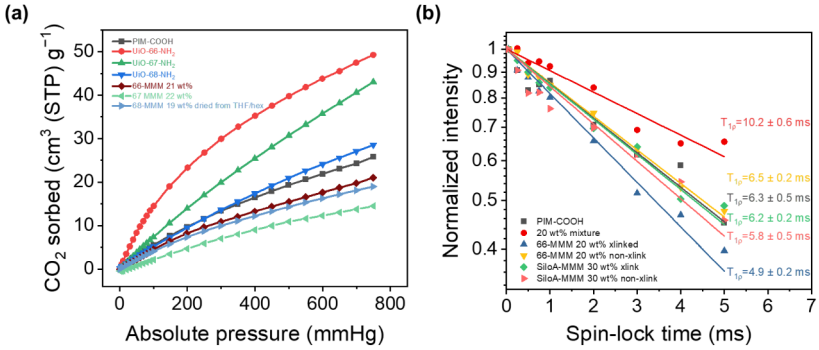

Figure 5. (a) CO_2 adsorption isotherms of PIM-COOH, UiO-6*x*-NH_2 MOFs, and MMMs. The UiO-6*x*-NH_2 samples were in powder form, while PIM-COOH and MMMs were freestanding films. UiO-66-NH_2 and UiO-67-NH_2 powders were dried from THF, and UiO-68-NH_2 was dried from a THF/hexane cosolvent. (b) ^{1}H $T_{1\rho}$ relaxation of PIM-COOH obtained from solid-state NMR CP-MAS experiments by varying the spin-lock time for a PIM-COOH film, a 20 wt % UiO-66-NH_2 in PIM-COOH powder mixture, a cross-linked 20 wt % 66-MMM, a non-cross-linked 20 wt % 66-MMM, a cross-linked 30 wt % silica-NH_2/PIM-COOH MMM, and a non-cross-linked 30 wt % silica-NH_2/PIM-COOH MMM.

uptake observed for 67-MMM compared to 68-MMM could be attributed to the partial framework collapse of UiO-67-NH_2 during cross-linking. The partial collapse of UiO-67-NH_2 was also observed from the lower CO_2 uptake in the cross-linked MMM compared to the non-cross-linked film (Figure S16).

Solid-state NMR experiments with cross-polarization magic angle spinning (CP-MAS) were used to investigate the dynamics of PIM-COOH in various samples. ^{13}C-detected ^{1}H $T_{1\rho}$ spin−lattice relaxation times of PIM-COOH were obtained by fitting the decay of the peak area with the spin-lock time to a single exponential decay function. A detailed description of the NMR data collection is provided in Section S2.4. The samples tested included a PIM-COOH film, cross-linked and non-cross-linked 20 wt % 66-MMM films, a powder mixture of 20 wt % UiO-66-NH_2 in PIM-COOH, and cross-linked and non-cross-linked MMMs with 30 wt % amine-functionalized silica nanoparticles in PIM-COOH. The silica nanoparticles with a particle size of 100 nm were chosen as a nonporous filler to understand the role of filler porosity on polymer infiltration. Note that these particles have a BET surface area of 43.2 m^2 g^{-1} due to external surface adsorption. The 67- and 68-MMMs were excluded from these NMR experiments due to their structural instability, which could hinder the investigation of polymer infiltration.

Figure 5b shows that the powder mixture has the longest ^{1}H $T_{1\rho}$, which is expected since PIM-COOH chains were not in close proximity to the MOF particles nor the other polymer chains. The pristine PIM-COOH film and the non-cross-linked 66-MMM showed similar ^{1}H $T_{1\rho}$ relaxation times, indicating that the dynamics of PIM-COOH in these two systems were comparable and that, without cross-linking, the presence of MOF had no significant effects on PIM-COOH. Both the non-cross-linked and cross-linked MMM had lower ^{1}H $T_{1\rho}$ values than that of the UiO-66-NH_2/PIM-COOH powder mixture, suggesting that a tighter polymer packing led to faster ^{1}H $T_{1\rho}$ relaxation. In the cross-linked 66-MMM, PIM-COOH exhibited faster relaxation due to rapid spin exchange from PIM-COOH to the MOFs. When the nonporous silica filler was used for the MMM, cross-linking did not lead to a reduction in ^{1}H $T_{1\rho}$, suggesting that the ^{1}H $T_{1\rho}$ reduction seen upon 66-MMM cross-linking is attributed to polymer infiltration into the fillers. Duan et al. studied two MMM systems, UiO-66/poly(ethylene oxide) (PEO) and UiO-66/poly(vinylidene fluoride) (PVDF), and found that the stronger interaction between UiO-66 and PEO compared to UiO-66 and PVDF led to more significant infiltration in the UiO-66/PEO MMMs.[46] Here, cross-linking effectively enhances the polymer−MOF interaction, thereby inducing infiltration.

The permeation and CO_2 sorption findings confirmed that pore blockage was present in the MMM systems. The ^{1}H $T_{1\rho}$ measurements further indicated that the change in PIM-COOH chain dynamics occurred only upon cross-linking due to polymer infiltration. However, the extent of polymer infiltration into the MOF pores remains unclear. Duan et al. used 2D ^{1}H−^{13}C heteronuclear correlation (HetCor) spectroscopy to quantify the extent of polymer infiltration for a UiO-66/PEO MMM based on the spin exchange time when a polymer−MOF cross peak appeared.[46] Unfortunately, the same technique could not be applied to our system because the polymer peaks overlapped with the MOF peaks on the ^{1}H spectra, prohibiting the identification of a polymer−MOF cross peak.

To investigate if the challenges observed with the UiO-6*x*-NH_2/PIM-COOH MMMs were specific to the polymer matrix, we also prepared MMMs using a different polymer matrix, 6FDA-DAM:DABA (3:2). PXRD patterns revealed a loss of MOF crystallinity for the larger-pore MOFs in MMMs, and CO_2 sorption analysis indicated pore blockage in all 6FDA-DAM:DABA MMMs (Figure S19). A more detailed discussion on the 6FDA-DAM:DABA-based MMM can be found in Section S3.9. These results emphasize that the issues of MOF structural instability and pore blockage in MMMs are not specific to one polymer, highlighting the challenges of leveraging larger-pore MOFs for separations.

3. CONCLUSIONS

PIM-COOH-based mixed-matrix membranes incorporating UiO-6*x*-NH_2 (x = 6, 7, or 8) MOFs were prepared. The UiO-6*x*-NH_2 MOFs were successfully synthesized, and guidelines for precisely tuning the MOF particle size were developed. Key challenges in activating larger-pore MOFs were identified, revealing that lower surface tension solvents more effectively activated UiO-68-NH_2. Consequently, the casting solvent for the UiO-68-NH_2/PIM-COOH MMMs was adjusted. A strong polymer−MOF interaction proved crucial for maintaining the structural integrity of MMMs in solvent environments. However, larger-pore MOFs experienced partial collapse during the cross-linking reaction, as evidenced by PXRD measurements of the cross-linked 67-MMMs. The OSN

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performance of MMMs was evaluated through pure-solvent permeation and dye rejection tests. The 66-MMM exhibited increased solvent permeability with higher MOF loading while maintaining a rejection profile similar to that of PIM-COOH. In contrast, the 67-MMM and 68-MMM showed similar or lower permeability compared to PIM-COOH, indicating partial pore collapse and blockage. Incorporating larger-pore MOFs had minimal effects on the MWCO due to the structural instability of the MOFs and nonideal polymer−MOF interfaces, which hindered access to the intrinsic properties of the MOFs. Further investigation of pore blockage was conducted through CO_2 sorption and solid-state NMR experiments. All PIM-COOH-based MMMs showed a lower CO_2 uptake than their parent materials, suggesting pore blockage. The ^1H $T_{1\rho}$ measurements on the UiO-66-NH_2/PIM-COOH system confirmed that polymer infiltration occurred in the cross-linked MMM samples. Overall, this study underscores the main challenges in OSN MMMs, such as MOF instability and nonideal polymer−MOF interfaces, providing valuable insights for the design and optimization of MMMs to address challenging separations.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge athttps://pubs.acs.org/doi/10.1021/acs.iecr.5c00174.

Methods (synthesis of bpdc-NH_2, tpdc-NH_2, UiO-67-NH_2, and UiO-68-NH_2; membrane fabrication); characterization (chemical and physical characterization methods, permeation test methods); results (characterization of bpdc-NH_2 and tpdc-NH_2; PXRD data of MOFs and MMMs; SEM images of thin films; confirmation of polymer−MOF cross-linking; stability of MOFs in EDC/NHS reaction; permeance values and performance benchmarking; CO_2 sorption; solid-state NMR; using 6FDA-DAM:DABA as the polymer matrix for MMMs) (PDF)

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W.-N.W.: conceptualization, methodology, formal analysis, investigation, data curation, writing − original draft. T.H.L.: conceptualization, methodology. A.H.: investigation, formal analysis. G.D.: investigation, formal analysis. Z.P.S.: conceptualization, writing − review and editing, supervision.

Notes

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■ ABBREVIATIONS

MOF, metal−organic framework; MMM, mixed-matrix membrane; PIM, polymer of intrinsic microporosity; bdp, 1,4-benzenedicarboxylate; bpdc, 4,4′-biphenyldicarboxylate; tpdc, terphenyl-4,4′-dicarboxylate; TEM, transmission electron microscopy; BET, Brunauer−Emmett−Teller; PXRD, powder X-ray diffraction; TGA, thermal gravimetric analysis; FT-IR, Fourier-transform infrared spectroscopy; SS NMR, solid-state nuclear magnetic resonance; THF, tetrahydrofuran; MeOH, methanol; IPA, isopropanol; ACN, acetonitrile

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